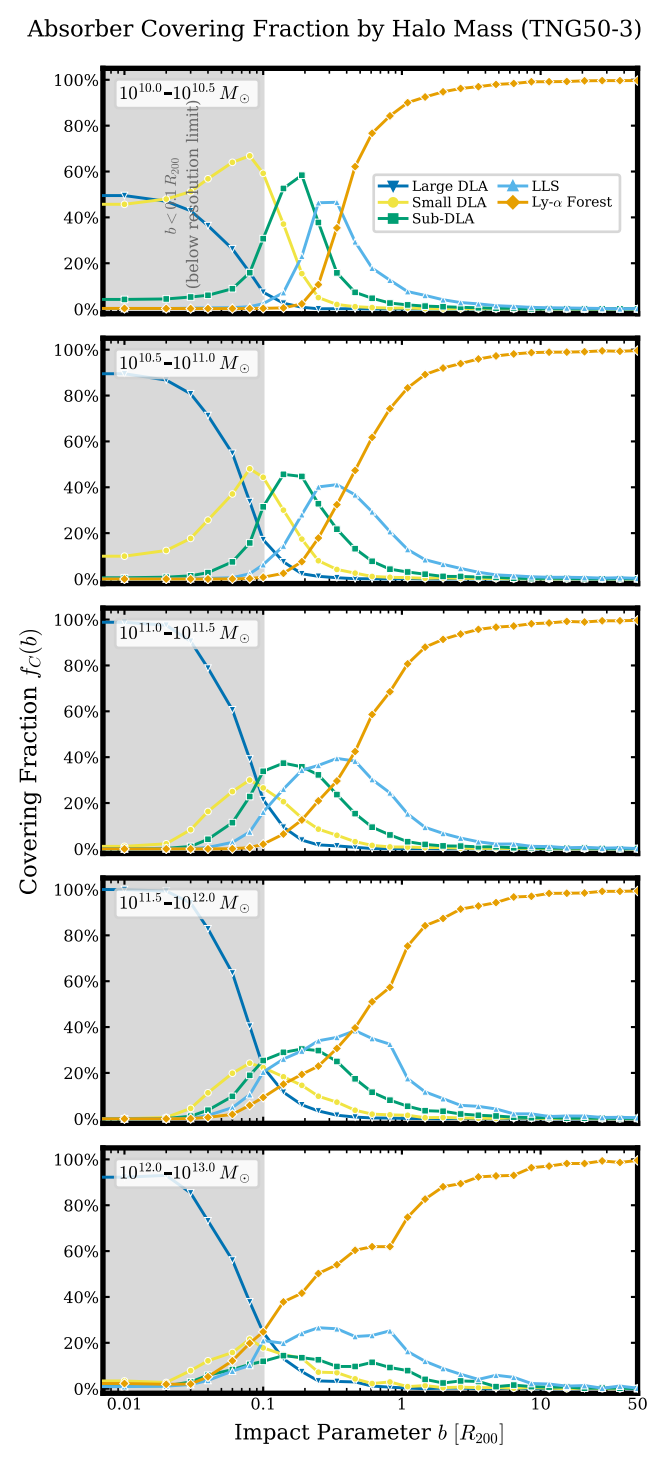

Absorber Covering Fraction by Halo Mass (TNG50-3)
$10^{10.0}$–$10^{10.5}\,M_\odot$
$10^{10.5}$–$10^{11.0}\,M_\odot$
$10^{11.0}$–$10^{11.5}\,M_\odot$
$10^{11.5}$–$10^{12.0}\,M_\odot$
$10^{12.0}$–$10^{13.0}\,M_\odot$
$b < 0.1\,R_{200}$
(below resolution limit)
Large DLA
Small DLA
Sub-DLA
LLS
Ly-α Forest
Covering Fraction $f_C(b)$
Impact Parameter b $[R_{200}]$
0%
20%
40%
60%
80%
100%
0.01
0.1
1
10
50